DigitalOcean → AWS EKS Full Cluster Migration Plan

Prepared: 2026-03-16 **Source:** DigitalOcean Kubernetes (DOKS) — 79 nodes, 183 namespaces, 1,141 pods **Target:** Amazon Elastic Kubernetes Service (EKS) **Application:** EMR-Bear Healthcare EHR Platform (Rails 5.2 + Puma)

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1. Dependency Graph & Migration Order

Service Dependency Tree (migrate bottom-up)

```
LAYER 0 — External (no migration needed, but config changes required)
├── AWS S3 (us-west-1, bucket: emrbear) ← already on AWS
├── AWS IoT Core (us-west-2) ← already on AWS
├── AWS DynamoDB (us-west-2) ← already on AWS
├── AWS Bedrock (us-west-2) ← already on AWS
├── Stripe API
├── Twilio API
├── Documo API
├── Change Healthcare API
├── Surescripts Network
├── Xero API
├── IDology API
├── Phaxio API
└── RXNT/NCPDP Gateway

LAYER 1 — Data Stores (migrate FIRST)
├── MySQL/MariaDB cluster (3 nodes: 10.138.255.216, 10.138.196.102, 10.138.76.186)
│   └── Per-tenant databases (~167 databases)
│   └── DelayedJob queue tables
│   └── CHC Lab Configuration
│   └── Reminder/Documo Configuration
├── MongoDB (mongodb.default.svc.cluster.local:27017)
│   └── bear3 database (forms, surveys, page content, configs)
├── Memcached (fallacious-bronco-memcached-{0,1})
│   └── Session cache, query cache (ephemeral — rebuild, don't migrate)
```

```
└── Redis 4.0.14 (bear-deployer namespace)
    └── Sidekiq queue for bear-deployer (rebuild, don't migrate)

LAYER 2 — Container Registry (migrate SECOND)
└── registry.emrbear.com (in-cluster Docker registry)
    └── Images: bear5:*, bear-deployer:*, bear-services:*, bear_x12:*, bear-xero:*
    └── Storage: DigitalOcean Spaces (sfo3)
    └── TARGET: AWS ECR

LAYER 3 — Shared Platform Services (migrate THIRD)
├── bear-services (services.emrbear.com)
│   ├── bear-services-nginx (reverse proxy)
│   ├── bear-services + bear-services-hds (API + HDS sidecar)
│   └── Depends on: MongoDB, MySQL
├── bear-x12 (parser.emrbear.com)
│   └── X12 EDI parser, 2 replicas
├── bear-xero (bear-xero.emrbear.com)
│   └── Xero accounting connector
├── trip-form-filler (3 replicas)
│   └── AHCCCS PDF form generation
├── bear-stats + bear-github (dash.emrbear.com)
│   └── Analytics dashboard + GitHub API proxy
├── bear-cron (daily + hourly CronJobs)
│   └── Scheduled maintenance tasks
└── Sentry (self-hosted at sentry.emrbear.com)
    └── NOTE: PVCs orphaned, may already be decommissioned

LAYER 4 — Prex Healthcare Services (migrate FOURTH)
├── cerner-web (2 replicas) → Drug/medication data
│   └── Consumed by: all tenant bear-app pods via PREX_DRUGS_HOST
├── production-surescripts-web (2 replicas) + worker
│   └── E-prescribing, consumed by tenant pods
├── staging-surescripts-web + worker
├── production-idology-web → Identity verification
├── staging-idology-web
├── saaspass-web → SSO/MFA
└── 4 CronJobs (Surescripts directory sync)

LAYER 5 — Tenant Applications (migrate LAST)
├── ~167 namespaces, each with:
│   ├── <tenant>-bear-app (Rails/Puma, 1-2 replicas)
│   └── delayed-<tenant>-bear-app-0 (DelayedJob worker, 1 replica)
├── Depends on: ALL of Layers 1-4
│   ├── MySQL (per-tenant DB via <tenant>-bear-db-secrets)
│   ├── MongoDB (via mongoid.yml in <tenant>-bear-app-secrets)
│   ├── Memcached (MEMCACHE_SERVERS env var)
│   ├── bear-services (BearConf.services_url)
│   ├── cerner-web (PREX_DRUGS_HOST)
│   ├── surescripts (via prex/service_endpoint.rb)
│   ├── idology (via prex/identities_controller.rb)
│   ├── saaspass (PREX_SAASPASS_HOST)
│   ├── trip-form-filler (trip_reporter_service.rb)
```

```
│   ├── bear-stats (bear_statistics/data_pusher.rb)
│   ├── bear-x12 (parse277.rb)
│   └── Shared secrets: shared-runtime-secrets, mariadb-certs
└── Special namespaces: gemini, demo5, demo6, demotat, staging
```

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2. Phase 0: Pre-Migration Preparation

0.1 Fix Pre-Existing Issues

- [ ] Resolve 3 failed Surescripts CronJobs (oldest: 491 days)
  - `production-surescripts-directory-full-sync-29397600` (Failed, 89d)
  - `production-surescripts-directory-nightly-sync-29373120` (Failed, 106d)
  - `staging-surescripts-directory-nightly-sync-28818900` (Failed, 491d)
- [ ] Investigate orphaned resources:
  - `latex-service` ClusterIP in `services` (no backing deployment)
  - `surescripts-web` ClusterIP in `prex` (no backing deployment)
  - Sentry PVCs (24Gi total, 0 running pods)
- [ ] Decommission legacy nginx-ingress v1 controller (`nginx-ingress-controller-manual`)
- [ ] Confirm `bear-deployer` (scaled to 0) and `production-gemini-web` (scaled to 0) disposition — migrate or drop?

0.2 Add Missing Resource Requests/Limits

- [ ] All `prex` namespace workloads have **zero** CPU/memory requests — profile with `kubectl top pods -n prex` and VPA recommendation mode
- [ ] All `services` namespace workloads are missing CPU **limits**
- [ ] Apply resource specs before migration so EKS node sizing is accurate

0.3 Audit External Partner IP Allowlists

Contact these partners about upcoming public IP change (DO → AWS):

- [ ] Surescripts (e-prescribing network) — TLS client cert may also be IP-bound
- [ ] Xero (accounting API)
- [ ] IDology (identity verification)
- [ ] Cerner (EHR)
- [ ] Change Healthcare (lab ordering)
- [ ] Phaxio (faxing)
- [ ] Documo (faxing)
- [ ] Mailgun (email — `system.emrbear.com` domain)

0.4 Inventory All Secrets

Export secret **names and keys** (not values) for all namespaces:

- [ ] 713 Opaque secrets
- [ ] 177 dockerconfigjson secrets (registry pull creds)
- [ ] Per-tenant: `<tenant>-bear-app-secrets` + `<tenant>-bear-db-secrets`
- [ ] Cluster-wide: `shared-runtime-secrets` , `mariadb-certs` (replicated ~167 times)

- [ ] Infrastructure: `bear-deployer-secrets`, `services-db-secrets`, `bear-xero-secrets`, `registry-secret`
- [ ] Prex: `cerner-secrets`, `idology-secrets`, `surescripts-secrets`, `surescripts-connection-cert`, `saaspass-secrets`, `prex-aws-secrets`, `prex-postgres-secrets`

0.5 Code Changes Required Before Migration

See Section 10 for the full list. Critical items:

- [ ] **SECURITY FIX**: Remove hardcoded X12 credentials from `app/lib/modules/parsers/x12/parse277.rb` (lines 61, 63) — move to environment variable or secret
- [ ] Extract all `*.svc.cluster.local` references to environment variables (they'll change if service names/namespaces change)
- [ ] Update HAProxy configs (`k8s/haproxy-config.yaml`, `k8s/haproxy-lb.yaml`, `k8s/haproxy.cfg`) with new database IPs
- [ ] Update CI/CD pipelines (`.drone.yml`, `.github/workflows/`, `Jenkinsfile`) to reference ECR instead of `registry.emrbear.com`

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3. Phase 1: AWS Foundation

1.1 EKS Cluster Provisioning

```
Kubernetes version: v1.32 (match current DOKS version)
Region: us-west-2 (co-locate with existing AWS services: IoT, DynamoDB, Bedrock, S3)
```

1.2 Node Groups

Node Group	Instance Type	Count	Purpose	Taints/Labels
`standard-pool`	`m6i.2xlarge` (8 vCPU / 32 GB)	40	Tenant bear-app pods	`pool=standard`
`delayed-job`	`m6i.xlarge` (4 vCPU / 16 GB)	38	DelayedJob workers	`pool=delayed-job`
`monitoring`	`t3.medium` (2 vCPU / 4 GB)	1	Prometheus, Grafana	`pool=monitoring`

- [ ] Reproduce node labels and taints from DO pools
- [ ] Enable cluster autoscaler or Karpenter

1.3 Networking

- [ ] VPC with private subnets (pods) + public subnets (load balancers)
- [ ] VPC peering or Transit Gateway to RDS/DocumentDB VPC if separate
- [ ] Security groups mirroring current NetworkPolicy rules:
  - Tenant pods: inbound 8080/80 only from ingress, bear-stats, bear-cron namespaces
  - No cross-tenant traffic

1.4 Storage Classes

DO Class	EKS Replacement	Provisioner

`do-block-storage`	`gp3` (default)	`ebs.csi.aws.com`
`do-block-storage-retain`	`gp3-retain`	`ebs.csi.aws.com`, reclaimPolicy: Retain
`do-block-storage-xfs`	`gp3-xfs`	`ebs.csi.aws.com`, fsType: xfs
`do-block-storage-xfs-retain`	`gp3-xfs-retain`	`ebs.csi.aws.com`, fsType: xfs, Retain

- [ ] Install AWS EBS CSI driver
- [ ] Create StorageClasses

1.5 Ingress Controller

- [ ] Install ingress-nginx v2 with AWS NLB (LoadBalancer service type)
- [ ] Configure TLS termination via ACM + cert-manager (currently TLS is terminated upstream)
- [ ] Single ingress class: `nginx`

1.6 Secrets Management

- [ ] Deploy AWS Secrets Manager CSI driver OR External Secrets Operator
- [ ] Create IAM roles for service accounts (IRSA) — replace all static AWS credentials:
  - `prex-aws-secrets` → IRSA
  - `AWS_ACCESS_KEY_ID` / `AWS_SECRET_ACCESS_KEY` in `shared-runtime-secrets` → IRSA
  - Velero cloud credentials → IRSA
- [ ] Provision all Opaque secrets into AWS Secrets Manager, organized by namespace

1.7 Container Registry (ECR)

- [ ] Create ECR repositories:
  - `bear5` (main app — production + develop tracks)
  - `bear-deployer`
  - `bear-services`
  - `bear-services-hds`
  - `bear_x12`
  - `bear-xero`
  - `bear-cron`
  - `bear-stats`
  - `bear-github`
  - `bear-gemini`
  - `curl` (used by CI)
- [ ] Mirror all images from `registry.emrbear.com` to ECR
- [ ] Verify `ghcr.io/emrbear/*` images (prex services, trip-reporter) pull correctly from EKS
- [ ] Create ECR pull secrets for each namespace (or use IRSA for ECR authentication)

1.8 Monitoring Stack

- [ ] Deploy kube-prometheus-stack (Grafana 9.x, Prometheus v2.39+, Alertmanager)
- [ ] Create 50Gi gp3 PVC for Prometheus TSDB
- [ ] Deploy `mysql-exporter` pointing to new RDS endpoint
- [ ] Migrate Grafana dashboards (export JSON from DO cluster)
- [ ] Configure Alertmanager rules

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4. Phase 2: Data Layer Migration

2.1 MySQL/MariaDB → AWS RDS

Current state: 3-node MariaDB cluster on DO droplets:

- Primary: `10.138.255.216:3306`
- Replica 1: `10.138.196.102:3306` (backup)
- Replica 2: `10.138.76.186:3306` (backup)
- Load balanced via HAProxy (in-cluster)
- ~167 tenant databases + infrastructure databases

Target: Amazon RDS for MariaDB (Multi-AZ)

Property	Value
Engine	MariaDB 10.6+ (compatible with current 10.2.29, test upgrade)
Instance class	`db.r6g.2xlarge` or larger (profile first)
Multi-AZ	Yes
Storage	gp3, sized to current data + 50% headroom
Encryption	At rest (KMS) + in transit (TLS)
Parameter group	Match current MariaDB tuning

Migration steps:

- [ ] Create RDS instance in same VPC as EKS
- [ ] Set up DMS (Database Migration Service) replication from DO MariaDB → RDS
- [ ] Validate all ~167 tenant databases replicate correctly
- [ ] Test application connectivity from EKS pods to RDS
- [ ] Update `<tenant>-bear-db-secrets` (`database.yml` , `.my.cnf`) to point to RDS endpoint
- [ ] Update HAProxy config or remove HAProxy entirely (RDS has built-in failover)
- [ ] Update `mariadb-certs` secret with new RDS TLS certificates (use Amazon RDS CA bundle)
- [ ] **CI/CD:** Update `BEAR_DATABASE_HOST` in `.drone.yml` (currently hardcoded to `10.138.56.0`)

2.2 MongoDB → Amazon DocumentDB (or self-hosted on EKS)

Current state: External droplet accessed via `mongodb.default.svc.cluster.local:27017` **Database:** `bear3` (forms, surveys, page content, configs)

Option A — Amazon DocumentDB (recommended):

- [ ] Create DocumentDB cluster (MongoDB 4.0 compatible)
- [ ] Use `mongodump` / `mongorestore` for migration
- [ ] Update `services-db-secrets` (`mongo-password`)
- [ ] Update `mongoid.yml` in `<tenant>-bear-app-secrets` to point to DocumentDB endpoint
- [ ] Update `BEAR_MONGO_HOST` environment variable

Option B — Self-hosted MongoDB on EKS:

- [ ] Deploy MongoDB StatefulSet with persistent volumes
- [ ] Not recommended — operational overhead

2.3 Prex PostgreSQL → Amazon RDS PostgreSQL

Current state: Referenced by `prex-postgres-secrets` and `prex-postgres-staging-secrets` **Used by:** cerner-web, surescripts, saaspass, idology

- [ ] Identify current PostgreSQL host (inspect secret values)
- [ ] Create RDS PostgreSQL instance
- [ ] Migrate data via `pg_dump` / `pg_restore` or DMS
- [ ] Update `prex-postgres-secrets` and `prex-postgres-staging-secrets`

2.4 Elasticsearch (Prex)

Current state: Referenced by `prex-elasticsearch-config` ConfigMap

- [ ] Identify current Elasticsearch host
- [ ] Migrate to Amazon OpenSearch Service or self-hosted on EKS
- [ ] Update `prex-elasticsearch-config`

2.5 Memcached (Ephemeral — Rebuild)

Current state: StatefulSet `fallacious-bronco-memcached-{0,1}` in `default` namespace

- [ ] Deploy Amazon ElastiCache for Memcached (2 nodes, matching current topology)
- [ ] OR deploy Memcached StatefulSet on EKS
- [ ] Update `MEMCACHE_SERVERS` env var in all tenant deployments
  - Current: `fallacious-bronco-memcached-0.fallacious-bronco-memcached.default.svc.cluster.local:11211:1,...`
  - New: ElastiCache endpoint or new StatefulSet DNS

2.6 Redis (Rebuild)

Current state: `redis-bear-deployer-master` StatefulSet in `bear-deployer` namespace (Bitnami Redis 4.0.14) **Also:** ActionCable requires Redis in production (`config/cable.yml` → `REDIS_URL`)

- [ ] Deploy Amazon ElastiCache for Redis or Redis StatefulSet on EKS
- [ ] Update `bear-deployer-secrets` (`REDIS_URL`)
- [ ] Update `REDIS_URL` for ActionCable in tenant pods (if not already in `shared-runtime-secrets`)

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5. Phase 3: Shared Infrastructure Services

3.1 `services` Namespace

Deploy in order:

1. **bear-services-nginx** + **bear-services** (+ bear-services-hds sidecar)
   - Update image refs to ECR

    - Update `services-db-secrets` with DocumentDB/MongoDB credentials
    - Verify `services.emrbear.com` ingress
    - Test: `curl https://services.emrbear.com/static/icd10cm_order_latest.txt` (static file serving)

2. **bear-x12** (2 replicas)
    - Update image ref to ECR
    - Update `bear-x12-secrets`
    - Verify `parser.emrbear.com` ingress
3. **bear-xero**
    - Update image ref to ECR
    - Update `bear-xero-secrets`, `xero-config`
    - Verify `bear-xero.emrbear.com` ingress
4. **trip-form-filler** (3 replicas)
    - Image from `ghcr.io/emrbear` — no registry change needed
    - Verify internal service at `trip-form-filler.services.svc.cluster.local:4567`
5. **docker-registry** — Decision point:
    - **Option A:** Migrate to ECR fully, decommission in-cluster registry
    - **Option B:** Redeploy on EKS pointing to S3 (instead of DO Spaces)
    - Recommendation: **Option A** — use ECR, update all CI/CD pipelines

3.2 `bear-stats` Namespace

- [ ] Deploy `bear-stats` + `bear-github`
- [ ] Update image refs to ECR
- [ ] Update `bear-github-secrets`
- [ ] Verify `dash.emrbear.com` ingress

3.3 `bear-cron` Namespace

- [ ] Deploy `bear-daily-job` CronJob (07:00 UTC)
- [ ] Deploy `bear-hourly-job` CronJob
- [ ] Update image refs to ECR

3.4 `bear-deployer` Namespace (if keeping)

- [ ] Deploy `redis-bear-deployer-master` StatefulSet
- [ ] Deploy `bear-deployer` + `bear-deployer-sidekiq`
- [ ] Update secrets, scale to desired replicas
- [ ] Verify `bear-deployer.emrbear.com` ingress

3.5 `monitoring` Namespace

- [ ] Deploy kube-prometheus-stack (see Phase 1.8)
- [ ] Deploy `mysql-exporter-m3` pointing to RDS

3.6 `velero` Namespace

- [ ] Deploy Velero v1.10+ with AWS plugin

- [ ] Configure S3 backup bucket
- [ ] Use IRSA instead of static cloud credentials

3.7 Sentry

- [ ] Decision: keep self-hosted `sentry.emrbear.com` or migrate to Sentry SaaS?
- [ ] If keeping: deploy Sentry Helm chart on EKS with Redis + Postgres
- [ ] If not: update all `SENTRY_DSN` values in `shared-runtime-secrets`

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6. Phase 4: Platform Services Migration

Validate all Layer 3 services are healthy before proceeding.

4.1 `gemini` Namespace

- [ ] Deploy `staging-gemini-web` + `staging-gemini-worker`
- [ ] Update `gemini-staging-secrets`
- [ ] Decide on `production-gemini-web` (currently scaled to 0)
- [ ] Verify `gemini-staging.emrbear.com` ingress

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7. Phase 5: Prex Healthcare Integrations

5.1 Prerequisites

- [ ] Prex PostgreSQL available on RDS (Phase 2.3)
- [ ] Prex Elasticsearch available on OpenSearch (Phase 2.4)
- [ ] Surescripts IP allowlist updated with AWS NAT Gateway IPs
- [ ] `surescripts-connection-cert` TLS client certificate verified/re-issued for new IPs

5.2 Deployment Order

1. **cerner-web** (2 replicas) — Drug data, consumed by all tenant pods
   - Secrets: `cerner-secrets`, `prex-aws-secrets` (→ IRSA), `prex-postgres-secrets`
   - Config: `cerner-config`
   - Test: `curl http://cerner-web.prex.svc.cluster.local/drugs/search_monographs?filename=d00148a1.htm`
2. **saaspass-web** — SSO/MFA
   - Secrets: `saaspass-secrets`, `prex-postgres-secrets`
   - Config: `saaspass-config`
3. **production-idology-web** + **staging-idology-web** — Identity verification
   - Secrets: `idology-secrets`
   - Config: `idology-config`
4. **production-surescripts-web** (2 replicas) + **production-surescripts-worker**
   - Secrets: `surescripts-secrets`, `surescripts-connection-cert`, `prex-aws-secrets`, `prex-postgres-secrets`
   - Config: `surescripts-config`

- Ingress: `surescripts-production.emrbear.com`

5. **staging-surescripts-web** + **staging-surescripts-worker**
   - Secrets: `surescripts-staging-secrets`, `surescripts-connection-cert`
   - Config: `surescripts-config-staging`
   - Ingress: `surescripts-staging.emrbear.com`
6. **CronJobs:**
   - `production-surescripts-directory-full-sync` (Sundays)
   - `production-surescripts-directory-nightly-sync` (Mon-Fri)
   - `staging-surescripts-directory-full-sync` (Saturdays)
   - `staging-surescripts-directory-nightly-sync` (Mon-Fri)

5.3 Validation

- [ ] Prescription creation end-to-end test
- [ ] Drug search via Cerner
- [ ] Identity verification via IDology
- [ ] SaaSPass authentication flow
- [ ] Surescripts directory sync runs successfully

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8. Phase 6: Tenant Application Migration

6.1 Per-Tenant Migration Checklist

For each of the ~167 tenant namespaces:

- [ ] Create namespace on EKS
- [ ] Deploy secrets:
  - `<tenant>-bear-app-secrets` (SECRET_KEY_BASE, master.key, aws.yml, mongoid.yml, symmetric-encryption keys)
  - `<tenant>-bear-db-secrets` (database.yml → RDS endpoint, .my.cnf → RDS endpoint)
  - `shared-runtime-secrets` (AWS_IOT_ENDPOINT, BEAR_GEMINI_URL, SENTRY_DSN)
  - `mariadb-certs` (RDS CA certificates)
  - `emrbear-registry-secret` (ECR pull secret or IRSA)
- [ ] Deploy ConfigMap: `<tenant>-bear-app-config` (custom_hosts)
- [ ] Deploy NetworkPolicy: `ingress-network-policy` (allow ingress from nginx-ingress, bear-stats, bear-cron)
- [ ] Deploy `<tenant>-bear-app` Deployment (1-2 replicas)
  - Image: ECR `bear5:<branch>-<sha>`
  - Env: `MEMCACHE_SERVERS` → new ElastiCache/Memcached endpoint
  - Verify: pods healthy, Rails boots, DB connection works
- [ ] Deploy `delayed-<tenant>-bear-app-0` Deployment (1 replica)
- [ ] Deploy Services: `<tenant>-bear-app` (ClusterIP:8080), `maintenance-page` (ClusterIP:80)
- [ ] Deploy Ingress: `<tenant>.emrbear.com` → `<tenant>-bear-app:8080`

6.2 Recommended Batch Strategy

Batch	Tenants	Purpose
Batch 0	`staging`, `demotat`, `demo5`, `demo6`	Test environments — validate full flow
Batch 1	5 low-traffic tenants on `develop` track	Canary production validation
Batch 2	20 tenants (smallest by user count)	Expand confidence
Batch 3	50 tenants	Bulk migration
Batch 4	Remaining ~90 tenants	Complete migration
Batch 5	`gemini`, `prex` staging/prod stragglers	Final cleanup

6.3 Per-Batch Validation

- [ ] Tenant app loads at `https://<tenant>.emrbear.com`
- [ ] User login works (session via Memcached)
- [ ] Patient record CRUD (MySQL)
- [ ] Document upload (S3)
- [ ] Prescription workflow (Surescripts → Prex)
- [ ] Fax send/receive (Twilio/Documo)
- [ ] DelayedJob background processing
- [ ] ActionCable WebSocket connectivity (Redis)

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9. Phase 7: DNS Cutover & Decommission

7.1 DNS Records to Update

Domain	Current Target	New Target
`*.emrbear.com` (wildcard)	DO LB `10.245.201.84`	AWS NLB/ALB IP or CNAME
`services.emrbear.com`	DO cluster	EKS ingress
`parser.emrbear.com`	DO cluster	EKS ingress
`bear-xero.emrbear.com`	DO cluster	EKS ingress
`dash.emrbear.com`	DO cluster	EKS ingress
`bear-deployer.emrbear.com`	DO cluster	EKS ingress
`registry.emrbear.com`	DO cluster	ECR (or decommission)
`sentry.emrbear.com`	DO cluster	EKS or Sentry SaaS
`surescripts-production.emrbear.com`	DO cluster	EKS ingress
`surescripts-staging.emrbear.com`	DO cluster	EKS ingress
`gemini-staging.emrbear.com`	DO cluster	EKS ingress
`assets.emrbear.com`	?	AWS CloudFront + S3

`help.emrbear.com`	?	Verify — may be external
`tainted.emrbear.com`	?	Verify — outside referral system
`system.emrbear.com`	Mailgun	Update SPF/DKIM if IP changes

7.2 Strategy

- **Option A (recommended):** Use weighted DNS (Route 53) to gradually shift traffic from DO → AWS
- **Option B:** Hard cutover with short TTL (riskier)

7.3 Decommission DO Cluster

- [ ] Verify zero traffic hitting DO cluster (monitor nginx-ingress access logs)
- [ ] Take final backups via Velero
- [ ] Delete DO Kubernetes cluster
- [ ] Delete DO Droplets (MariaDB nodes, MongoDB) after RDS validation period
- [ ] Delete DO Spaces storage (registry data) after ECR migration confirmed
- [ ] Cancel DO infrastructure billing

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10. Hardcoded References Requiring Code Changes

CRITICAL — Security

File	Line	Issue	
`app/lib/modules/parsers/x12/parse277.rb`	61	Hardcoded credentials: `emr-bear:88163381e51c2b08fecf1147710c538fc79a36d1` in cluster-local URL	Mo env Ku sec
`app/lib/modules/parsers/x12/parse277.rb`	63	Same credentials in `parser.emrbear.com` URL	Mo env Ku sec

HIGH — Kubernetes Service DNS (*.svc.cluster.local)

These are defaults that fall back when env vars are not set. If namespaces and service names remain identical on EKS, **no code change is needed**. But they should be externalized to env vars for portability:

File	Line	Reference	
`app/lib/prex/medication/base.rb`	28	`cerner-web.prex.svc.cluster.local`	P e
`app/services/prex/drug/create.rb`	38	`cerner-web.prex.svc.cluster.local`	P e
`app/services/prex/supply/create.rb`	49	`cerner-web.prex.svc.cluster.local`	P e

app/services/prex/consents/request_monograph.rb	26	cerner-web.prex.svc.cluster.local	P e
app/controllers/prex/identities_controller.rb	152	idology-web.prex.svc.cluster.local	N (
app/lib/prex/saaspass/base.rb	47	saaspass-web.prex.svc.cluster.local	P
app/models/fund/invoice/remote_client.rb	2	bear-xero.services.svc.cluster.local:4567	N (
app/lib/modules/parsers/x12/parse277.rb	61	bear-x12.services.svc.cluster.local:3000	N
app/services/trip_reporter_service.rb	4	trip-form-filler.services.svc.cluster.local:4567	N
app/classes/bear_statistics/data_pusher.rb	47-48	bear-stats.bear-stats.svc.cluster.local:3000	B e
app/controllers/releases_changelog_controller.rb	27	bear-stats.bear-stats.svc.cluster.local:3000	N
app/classes/bear_conf.rb	23-24	bear-services.services.svc.cluster.local	P c s e

MEDIUM — Hardcoded DigitalOcean IPs

File	Line	IP	Purpose	Fix
k8s/haproxy-config.yaml	34-36	10.138.255.216, 10.138.196.102, 10.138.76.186	MariaDB cluster nodes	Replace with RDS endpoint
k8s/haproxy-lb.yaml	41-43	Same 3 IPs	MariaDB load balancer	Replace with RDS or remove HAProxy
k8s/haproxy.cfg	25-27	Same 3 IPs	MariaDB config	Replace with RDS or remove HAProxy
.drone.yml	41	10.138.56.0	CI database host	Update to CI database endpoint
.drone.yml	86, 102	10.138.192.109	K8s API via HAProxy	Update to EKS API endpoint

`app/lib/modules/billing/magellan.rb`	31-32	`10.138.108.116:1080`	SOCKS proxy for Magellan billing	Update to new proxy or direct route

MEDIUM — CI/CD Pipeline Updates

File	Lines	Issue	Fix
`.drone.yml`	16-17, 37, 82, 98	`registry.emrbear.com` image refs	Update to ECR
`.github/workflows/deploy_production.yml`	48-52, 74	`registry.emrbear.com/bear5`	Update to ECR
`.github/workflows/deploy_staging.yml`	38, 40, 145	`registry.emrbear.com`	Update to ECR
`Jenkinsfile`	10, 61, 76, 146-147, 164	`registry.emrbear.com`, `bear-deployer.emrbear.com`	Update to ECR + new deployer URL

LOW — Domain References (functional, don't need code changes)

These use `*.emrbear.com` public domains that will continue to work after DNS cutover:

- `config/environments/production.rb:33` — `assets.emrbear.com` (asset host)
- `config/environments/production.rb:95-96` — `system.emrbear.com` (Mailgun)
- `app/services/prex/service_endpoint.rb:32` — `surescripts-{env}.emrbear.com`
- `app/classes/bear_conf.rb:27` — `services.emrbear.com` (external fallback)
- `app/classes/bear_statistics/data_pusher.rb:51` — `dash.emrbear.com`
- `app/classes/outside_referral.rb:120,128` — `tainted.emrbear.com`
- Fax callback URLs using `BearConf.subdomain` — dynamic, will resolve correctly
- `config/initializers/cors.rb:4` — `emrbear.com` CORS origin ✅
- Static file rake tasks referencing `services.emrbear.com/static/*` ✅

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11. Risk Register

#	Risk	Severity	Mitigation
1	Database migration data loss/corruption	**Critical**	Use DMS with continuous replication; validate row counts + checksums per tenant DB
2	Surescripts connectivity failure (IP change + TLS cert)	**Critical**	Pre-coordinate with Surescripts; test in staging first; have rollback DNS ready
3	In-cluster registry unavailable during transition	**High**	Mirror all images to ECR before starting; keep DO registry running until all pods on EKS pull from ECR

4	Secret values lost/mismatched during re-provisioning	**High**	Script secret export (values) from DO; import to AWS Secrets Manager; automate verification
5	Cross-service DNS resolution failures	**High**	Keep identical namespace + service names on EKS; test all `*.svc.cluster.local` references
6	Memcached cache invalidation causing session drops	**Medium**	Coordinate cutover during low-traffic window; warn users of potential re-login
7	Change Healthcare IP allowlist not updated	**Medium**	Contact CHC before migration; use NAT Gateway with static Elastic IPs
8	DelayedJob queue backlog during migration	**Medium**	Drain DelayedJob queue before tenant cutover; monitor `delayed_jobs` table
9	MongoDB/DocumentDB compatibility issues (Mongoid 7.0.2)	**Medium**	Test DocumentDB compatibility mode; Mongoid 7.x supports MongoDB 4.0 API
10	Magellan billing SOCKS proxy unreachable	**Medium**	Identify proxy location; re-provision on AWS or establish VPN back to DO network

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12. Rollback Plan

Per-Phase Rollback

Phase	Rollback Action	RTO
Phase 1 (AWS Foundation)	Delete EKS cluster + resources	Clean, no impact
Phase 2 (Data Layer)	Stop DMS replication; DO databases remain primary	Minutes
Phase 3-5 (Services)	DNS still points to DO; EKS services are secondary	No user impact
Phase 6 (Tenants)	Revert DNS to DO LB IP; DO pods still running	Minutes (DNS TTL)
Phase 7 (DNS Cutover)	Revert DNS records to DO endpoints	DNS TTL (keep low: 60s during cutover)

Key Principle

Run both clusters in parallel until EKS is fully validated. DO cluster remains the primary until DNS cutover. Database replication runs DO → AWS (DMS), so DO always has the latest data until final cutover.